

Al-Aqsa University
Faculty of Computers and IT
Dept. of Info & Computer science



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم علم الحاسوب والمعلومات

Advanced Database

Assignment 3

Presented to: Eng. Maha AL-Atteli

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Semester: First Semester 2025/2026

Question 1 : Write a function that print out the maximum and minimum salary of employees in a specific job.)

The Function:

```
1  CREATE OR REPLACE FUNCTION GET_JOB_SALARY_RANGE
2  (p_job IN EMP.JOB%TYPE)
3  RETURN VARCHAR2
4  IS
5      CURSOR C1 IS
6          SELECT MIN(SAL) AS MIN_SAL,
7                 MAX(SAL) AS MAX_SAL
8          FROM EMP
9          WHERE JOB = p_job;
10
11      V_MIN EMP.SAL%TYPE;
12      V_MAX EMP.SAL%TYPE;
13  BEGIN
14      FOR REC IN C1 LOOP
15          V_MIN := REC.MIN_SAL;
16          V_MAX := REC.MAX_SAL;
17
18          DBMS_OUTPUT.PUT_LINE('JOB TITLE: ' || p_job);
19          DBMS_OUTPUT.PUT_LINE('MINIMUM SALARY: ' || V_MIN);
20          DBMS_OUTPUT.PUT_LINE('MAXIMUM SALARY: ' || V_MAX);
21      END LOOP;
22
23      RETURN V_MIN || ' - ' || V_MAX;
24  END;
```

The Result:

Language: SQL ? Rows: 10

```
1 CREATE OR REPLACE FUNCTION GET_JOB_SALARY_RANGE
2 (p_job IN EMP.JOB%TYPE)
3 RETURN VARCHAR2
4 IS
5     CURSOR C1 IS
6         SELECT MIN(SAL) AS MIN_SAL,
7                MAX(SAL) AS MAX_SAL
8         FROM EMP
9         WHERE JOB = p_job;
10
11     V_MIN EMP.SAL%TYPE;
12     V_MAX EMP.SAL%TYPE;
13 BEGIN
14     FOR R IN C1 LOOP
```

Results Explain Describe Saved SQL History

Function created.

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```
25
26 BEGIN
27     DBMS_OUTPUT.PUT_LINE( GET_JOB_SALARY_RANGE('CLERK') );
28 END;
29
```

Results Explain Describe Saved SQL History

JOB TITLE: CLERK
MINIMUM SALARY: 800
MAXIMUM SALARY: 1300
800 - 1300

Statement processed.

Question 2 : Create procedure that print out how many employees and average salary exist in a specific department

The Procedure:

```
31 CREATE OR REPLACE PROCEDURE DEPT_STATS
32 (p_deptno IN EMP.DEPTNO%TYPE)
33 IS
34     CURSOR C2 IS
35         SELECT COUNT(*) AS EMP_COUNT,
36                AVG(SAL) AS AVG_SAL
37         FROM EMP
38         WHERE DEPTNO = p_deptno;
39
40     V_COUNT NUMBER;
41     V_AVG   NUMBER;
42 BEGIN
43     FOR REC IN C2 LOOP
44         V_COUNT := REC.EMP_COUNT;
45         V_AVG   := REC.AVG_SAL;
46
47         DBMS_OUTPUT.PUT_LINE('DEPARTMENT NUMBER: ' || p_deptno);
48         DBMS_OUTPUT.PUT_LINE('NUMBER OF EMPLOYEES: ' || V_COUNT);
49         DBMS_OUTPUT.PUT_LINE('AVERAGE SALARY: ' || V_AVG);
50     END LOOP;
51 END;
```

The Result:

```

31 CREATE OR REPLACE PROCEDURE DEPT_STATS
32 (p_deptno IN EMP.DEPTNO%TYPE)
33 IS
34     CURSOR C2 IS
35         SELECT COUNT(*) AS EMP_COUNT,
36                AVG(SAL) AS AVG_SAL
37         FROM EMP
38        WHERE DEPTNO = p_deptno;
39
40     V_COUNT NUMBER;
41     V_AVG   NUMBER;
42 BEGIN
43     FOR REC IN C2 LOOP

```

Results Explain Describe Saved SQL History

Procedure created.

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```
53 BEGIN
54   DEPT_STATS(10);
55 END;
```

Results Explain Describe Saved SQL History

[illegible]

Conclusion:

In this assignment, I applied the main PL/SQL concepts learned this week by creating a function and a procedure that work with the EMP table. The function returned the minimum and maximum salary for a specific job, while the procedure printed the number of employees and the average salary for a selected department. I used parameters, %TYPE, cursors, loops, and DBMS_OUTPUT to process and display the results. This work helped me understand how PL/SQL programs interact with database tables and how to execute stored programs inside Oracle APEX.

"واخر دعواهم ان الحمد لله رب العالمين"